

WASP-11b

R-0683

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_251031 · created 2026-07-27T19:48:45+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460979.9471 +/- 0.0039 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.176 +/- 0.052
Transit depth (Rp/Rs)^2	3.1 +/- 1.82 [%]
Orbital Inclination (inc)	87.88 +/- 1.71
Airmass coefficient 1 (a1)	1.83 +/- 1.18
Airmass coefficient 2 (a2)	-0.57 +/- 1.4
Scatter in the residuals of the lightcurve fit is	58.25 %
Best Comparison Star	None
Optimal Aperture	1.7
Optimal Annulus	5.0
Transit Duration (day)	0.061 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.176 ± 0.052 vs 0.131 (deviation 0.87σ)
Duration fit vs published	0.061 d vs 0.1075 d (deviation 4.23σ)
Mid-time O–C	-4.3 min
Depth SNR	3.4
Residual scatter	58.25 %

Light curve

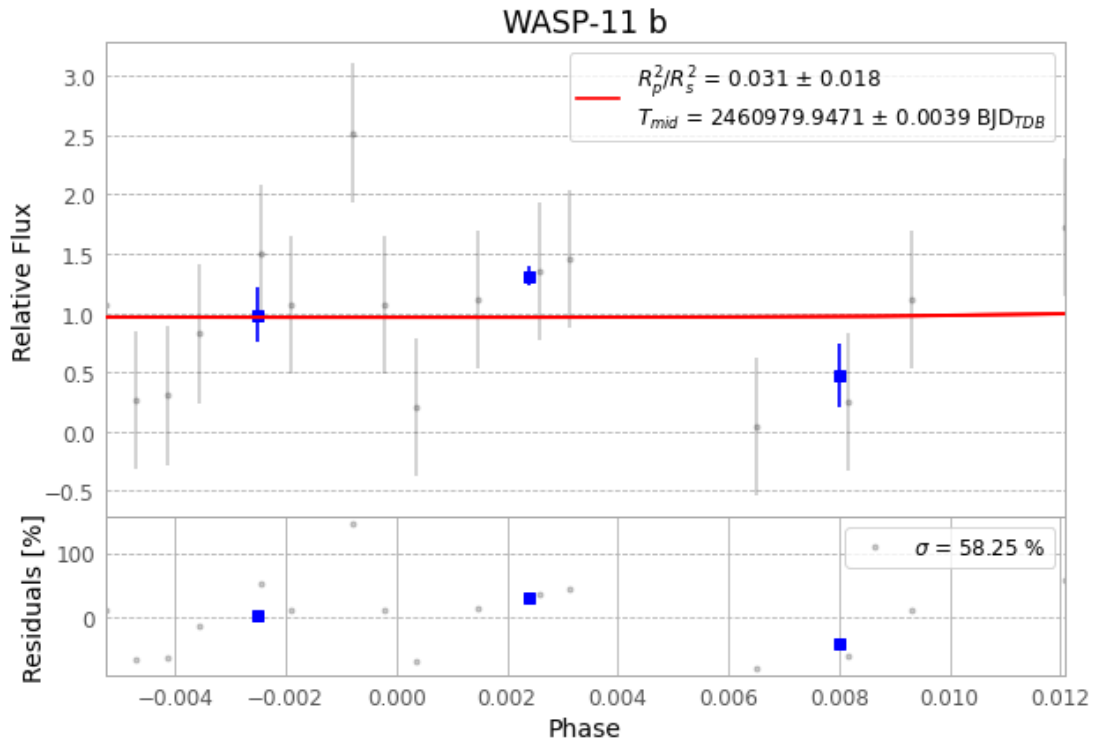


Figure 1 — FinalLightCurve_WASP-11 b_2025-10-31.png (SHA-256 914608dae13be343...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [390, 242], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fb09b59a6a64485b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813

Data provenance

31 FITS frames (20 MB), HATP-10251031100315.FITS → HATP-10251031113915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-251031.log (34c70ae3186b...), FinalLightCurve_WASP-11 b_2025-10-31.png (914608dae13b...), FinalLightCurve_WASP-11 b_2025-10-31.csv (7fa7c49a1dc3...)

Compute resources

Wall time 90.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 19:48Z.